Product Backlog Refinement Activity Task Sheet

Session Duration: 30–45 Minutes
Team Structure: Groups of 6 Developers
Your Role: Development Team (Software Engineers / UI Designers)
Instructor Role: Product Owner (PO)

1. Project Brief: Campus Express (Food Delivery App)

Campus Express is a mobile application designed for university students and staff to order meals from campus dining halls and local food vendors for fast delivery directly to dorms, libraries, or lecture halls.
As the Product Owner (PO), I understand the business vision and user needs, but **I rely on you—the Development Team—to evaluate technical feasibility, highlight implementation complexities, ask clarifying questions, and help determine the final release priority.**

2. Session Objectives

During this refinement session, your group will work through the initial Product Backlog to:

1. **Clarify Requirements:** Ask the PO questions to uncover missing edge cases or technical assumptions.
2. **Estimate Relative Effort (Sizing):** Assign Story Points using the **Modified Fibonacci Sequence** [1, 2, 3, 5, 8, 13, 20, 40, 100] based on Complexity, Effort, and Risk/Uncertainty.
3. **Decompose Large Items:** If a story is too large or risky (typically >13 points), split it into smaller, manageable PBIs.
4. **Advise on Priority:** Help the PO adjust the backlog priority based on technical dependencies and effort vs. value tradeoffs.

3. Definition of Done (DoD) Reference

Keep our baseline quality standard in mind when estimating effort:

- [ ] UI wireframe sketched with interactive navigation flow.
- [ ] Acceptance criteria fully satisfied.
- [ ] Peer-reviewed by at least one other developer.
- [ ] Verified integration with upstream/downstream screens.

4. Initial Product Backlog (10 User Stories)

ID	User Story	Questions for PO	Initial Estimate (Points)	Refined Estimate / Split PBIs	Suggested Priority
US-01	**User Authentication:** As a student, I want to view a login/register screen with email and password fields so that I can access my campus account.				
US-02	**Restaurant Browsing:** As a hungry student, I want to see a scrollable list/grid of open campus eateries with ratings and delivery times so that I can pick a place to eat.				
US-03	**Menu Selection:** As a user, I want to view a restaurant's menu with item descriptions and price tags so that I can select my food items.				
US-04	**Customization Modal:** As a user, I want a pop-up modal when tapping a dish to select portion size and add special instructions (e.g., "no onions").				
US-05	**Cart Summary:** As a user, I want a shopping cart view displaying selected				

	items, subtotal, delivery fee, and total price so that I can review my order.				
US-06	**Checkout & Payment:** As a user, I want a checkout screen with delivery location selection (e.g., Library, Dorm A) and payment method selection (e.g., Campus Card, Credit Card).				
US-07	**Live Order Tracking:** As a customer, I want a progress screen showing order stages (Received > Preparing > Out for Delivery > Arrived) so that I know when to collect my food.				
US-08	**Search & Filter:** As a user, I want a top search bar with filter tags (e.g., "Vegetarian", "Under $10", "Fastest") so that I can quickly find specific foods.				
US-09	**Ratings & Reviews:** As a user, I want to view customer star ratings and text reviews on a restaurant's detail page so that I can judge food quality.				
US-10	**Order History:** As a returning user, I want a profile tab showing past orders with a "Reorder" button so that I can quickly repurchase frequent meals.				

5. Step-by-Step Instructions for Teams

- **Step 1: Read & Question (10 Mins)**
  Review each story as a group. Formulate specific technical or business questions for the Product Owner (e.g., *"Does Checkout need real-time credit card gateway integration or just campus card mockups?"*). Raise your hand to ask the PO during the session.
- **Step 2: Relative Sizing (10 Mins)**
  Run rapid **Planning Poker** within your team:
  - Pick **US-01** or **US-03** as your baseline reference task (e.g., 2 or 3 Story Points).
  - Vote on remaining items using cards [1, 2, 3, 5, 8, 13, 20, 40, 100].
  - If estimates diverge significantly, have the highest and lowest voters explain their technical reasoning, then re-vote until consensus is reached.
- **Step 3: Backlog Decompositions (10 Mins)**
  Identify any story estimated at **13 points or higher**. Split it into two or more smaller user stories (e.g., splitting **US-06 Checkout & Payment** into *Delivery Location Selection* and *Payment Gateway Processing*).
- **Step 4: Priority Alignment & Delivery (5 Mins)**
  Re-order your refined backlog table based on technical dependencies (e.g., Menu Selection must precede Cart Summary) and communicate recommended priority adjustments to the PO.